Markscheme

Mathematics: analysis and approaches
Practice question bank

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92. (a)

Consider the statement: "$x^2 \geq x$ for every real number x." This statement is false.

Show that $x = \dfrac{1}{2}$ is a counter-example, and hence evaluate $x^2 - x$ at this value.

at $x = \frac{1}{2}$: $x^2 = \frac{1}{4}$, which is less than $x = \frac{1}{2}$, so the statement fails *(M1)* **R1**

$$x^2 - x = \frac{1}{4} - \frac{1}{2}$$ *(M1)*

$$= -0.25$$ **A1**

[3 marks]